

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B Tech (Electronics and Communication) Semester-III Examination Nov-2010(Regular)
EC – 201: Digital Electronics & Logic Design

Date: 26/11/2010, Friday

Time: 10:30 a.m to 1:30 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in separate answer books.
3. Figures to the right indicate *full* marks.

FOR REFERENCE

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|-----------|--|---|
| Q-1 (a) | Define the following terms :
Propagation Delay, Fan out | 2 |
| (b) | Write the first 20 decimal digits in base 3 | 3 |
| (c) | Convert following number in to decimal (1001001.011) ₂ | 2 |
| (d) | Subtract the following decimal number using 10's complement. 753-864 | 3 |
| (e) | Covert the hexadecimal 2AC5.D to decimal | 1 |
| Q-2 (a) | Using Boolean algebra prove the following
1) $(B+BC)(B+B'C)(B+D)=B$
2) $XYZ+X'Y+XYZ'=Y$ | 4 |
| (b) | Convert the following number in other canonical form
1) $F(x,y,z)=\sum(1,3,7)$
2) $F(x,y,z)=\prod(0,3,6,7)$ | 4 |
| (c) | Design a combinational circuit to check for even parity of three bits. A logic 1 output is required when the three bits do not constitute an even parity | 4 |
| OR | | |
| Q-2 (a) | Simplify the following Boolean function using K map
$F(A,B,C,D)=\sum(7,13,14,15)$ | 3 |
| (b) | Simplify the following Boolean function using the tabulation method
$F=\sum(0,1,2,8,10,11,14,15)$ | 6 |
| (c) | Obtain the simplified expression in sum of product of following function USING K map. $D(A'+B)+B'(C+AD)$ | 3 |
| Q-3 (a) | Express following function in sum of min terms:
1) $F(A,B,C)=(A'+B)(B'+C)$
2) $F(x,y,z)=(xy+z)(y+xz)$ | 4 |
| (b) | Implement the following Boolean function with logic diagram only with OR and NOT gates
1) $F=xy+x'y'+y'z$ | 4 |

$$2) F=y(wz'+wz)+xy$$

- (c) Obtain truth table for $F=xy+xy'+y'z$ 1
- (d) Show the dual of exclusive- or is equal to its complement 3
- OR
- Q- 3 (a) Draw logical diagram of following Boolean function only with NAND gates 4
(with two inputs)
 $(AB+A'B')(C'D+CD')$
- (b) Simplify the following Boolean function using don't care condition in sum of product(K map) 4
 $F=ACE+A'CD'E'+A'C'DE$
 $d=DE'+A'D'E+AD'E'$
- (c) Show how a full adder can be converted to a full subtractor with the addition of one inverted circuit 4

Section II

- Q- 4(a) What is the difference between sequential circuit and combinational circuit? 2
- (b) Give the difference between synchronous and asynchronous counter 2
- (c) Define the following terms with respect to sequential circuit. 2
a) state table , b) excitation table
- (d) What is raised around condition in j-k flipflop 2
- (e) Differentiate between decoder and multiplexer. How decoder can be used as multiplexer? 3
- Q- 5(a) Design a BCD to excess 3 code converter with a BCD to decimal decoder and four OR gates. 6
- (b) Design a 4 line to 2 line priority encoder. Include an output E to indicate that at least one input is a 1. 6

OR

- Q-5 (a) Write the difference between serial and parallel transfer? 4
Explain serial transfer using example
- (b) Design the sequential circuit whose state table is listed as follows 4

Present state		input	Next states		output
A1	A2	X	A1	A2	Y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	0	0	1
1	0	0	1	0	0
1	0	1	0	1	0
1	1	0	1	1	0
1	1	1	0	0	1

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|-----|--|----|
| (C) | Implement a full adder circuit with multiplexers. | 4 |
| Q-6 | Attempt any three | 12 |
| (a) | Design a counter with following binary sequence : 0,1,,2,3,4,5,6,7 and repeat.
Use RS flip flops. | |
| (b) | Show logic diagram of a clocked D flip flop with AND and NOR gates. | |
| (c) | Write technical note on Johnson counter | |
| (d) | Write technical note on look ahead carry generator. | |
| (e) | Draw the logic diagram (showing all gates) of a master slave D flip flop. Use
NAND gates | |